Joint embedding for finite order forests: a decision theorem and a width classification

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Abstract

Given finitely many forbidden finite order forests, can one decide whether any two permitted forests have a permitted common extension? We answer this question effectively by passing to comparability graphs and applying Carter's decision theorem for hereditary graph classes with a forbidden induced four-vertex path. The transfer respects existence of embeddings, although a particular graph embedding need not preserve the order. We also prove an explicit classification: forests of width at most m have the joint embedding property exactly when $m \leq 3$ (with width zero included if the empty forest is allowed). The negative cases have finite certificates, and the positive cases yield countable forests with the prescribed age. The general algorithm is an application of Carter's published preprint; the order-theoretic transfer and the width argument are given in full.

1 The question and the role of the graph reduction

An order forest is a partially ordered set in which the predecessors of each element form a chain. A finite forest can have several minimal elements. An embedding preserves and reflects the order. In particular, it need not preserve a root, an immediate-predecessor relation, or a meet; none of these is part of the language.

For a finite family $\mathcal{U}$ of finite order forests, put

$$\mathcal{C}(\mathcal{U}) = \{T : T \text{ is a finite order forest and no } U \in \mathcal{U} \text{ embeds into } T\}.$$

A class has the *joint embedding property* (JEP) if any two members embed into a third member. Gonzalez, Rossegger and Turetsky ask which finite forbidden antichains have this property in Question 2 of [2]. Here an antichain of forbidden structures is under the embedding relation; it need not consist of poset antichains.

The question is algorithmic as well as structural. Failure of JEP can be witnessed by two finite forests, but searching for such a pair does not decide the positive case. The missing positive termination guarantee comes from Carter's graph theorem [1]. The main task in using that theorem is to justify the translation of embeddings, rather than merely recognize a familiar graph class.

As a first example, forbidding a two-element antichain leaves just chains. Two finite chains embed into a sufficiently long chain, so JEP holds. Forbidding a five-element antichain instead allows width four. We shall see that a balanced four-leaf binary tree and a four-leaf comb have no common extension of width four. Thus the obstruction is branching shape, even though the embeddings do not name branching operations.

2 Recovering the order from its comparability graph

Write $\mathrm{Comp}(T)$ for the simple graph on T in which distinct vertices are adjacent exactly when they are comparable. Write P_4 for the four-vertex path and C_4 for the four-vertex cycle. Graph embeddings below are induced embeddings. For a graph family $\mathcal{F}$, $\mathrm{Forb}_{\mathrm{ind}}(\mathcal{F})$ denotes the finite graphs avoiding every member of $\mathcal{F}$ as an induced subgraph.

Lemma 2.1. *The comparability graphs of finite order forests are precisely* $\mathrm{Forb}_{\mathrm{ind}}(P_4, C_4)$. *Each such graph determines its order forest up to isomorphism. Moreover, for finite order forests* S, T,

$$S \hookrightarrow T \quad \Longleftrightarrow \quad \mathrm{Comp}(S) \hookrightarrow \mathrm{Comp}(T).$$

Proof. Each connected component of a finite forest has a least element: the root beneath a vertex is unchanged along any edge of its comparability graph, and hence along a connected component. Its graph is obtained by adding a universal vertex to the disjoint union of the graphs above that root. Recursion excludes induced P_4 and C_4: each forbidden graph is connected and neither has a universal vertex.

Conversely, a nonempty connected graph avoiding P_4, C_4 has a universal vertex. Choose v with maximal size of its closed neighborhood $N[v]$. If v is not universal, a shortest path to a nonneighbor has length two, since a longer shortest path would contain an induced P_4. Call the path v, w, x. Because $x \in N[w] \setminus N[v]$ and $|N[v]| \geq |N[w]|$, choose $u \in N[v] \setminus N[w]$. The four distinct vertices u, v, w, x induce P_4 when ux is absent and C_4 when it is present. Both are impossible. Remove a universal vertex, recursively represent the remaining components, and place the removed vertex below them. Represent a disconnected graph by a disjoint union.

For uniqueness, all universal vertices in a connected representing forest form an initial chain. They are pairwise comparable. If $z \leq u$ and u is universal, then z is comparable with every vertex: use transitivity above u, and linearity of the predecessors of u below it. Thus z is universal too. Every nonuniversal vertex lies above the whole universal chain. Deleting that chain leaves the uniquely specified graph components. Induction reconstructs their orders up to isomorphism. Permuting the names of the universal vertices does not change that isomorphism type.

An order embedding gives an induced graph embedding. For the converse, give the image of a graph embedding the order induced from T. It is a forest whose comparability graph is isomorphic to $\mathrm{Comp}(S)$. Uniqueness gives an order isomorphism from S to that image. □

The last argument asserts existence of an order embedding, not that the original graph map is an order embedding. For a two-element chain, the transposition is a graph automorphism but reverses the order. That distinction is harmless for JEP and essential for interpreting the reduction correctly. These graphs are commonly called trivially perfect graphs; the proof above supplies all properties needed here.

3 Deciding the forbidden-family question

We use the following published result in its stated form: Carter's Corollary 1.5 [1] decides JEP for a hereditary class of finite graphs specified by a finite forbidden induced-subgraph family containing P_4. We use this graph corollary, not his distinct notions of maps between rooted trees.

Theorem 3.1. *There is an algorithm which, given any finite family $\mathcal{U}$ of finite order forests, decides whether $\mathcal{C}(\mathcal{U})$ has JEP. More precisely, set*

$$\mathcal{F} = \{P_4, C_4\} \cup \{\mathrm{Comp}(U) : U \in \mathcal{U}\}.$$

Then $\mathcal{C}(\mathcal{U})$ has JEP if and only if $\mathrm{Forb}_{\mathrm{ind}}(\mathcal{F})$ has JEP.

Proof. Lemma 2.1 gives an effective correspondence on isomorphism types between all finite forests and $\mathrm{Forb}_{\mathrm{ind}}(P_4, C_4)$. It preserves and reflects existence of embeddings. In particular, T avoids every U exactly when $\mathrm{Comp}(T)$ avoids every $\mathrm{Comp}(U)$. A common extension on either side corresponds to a common extension on the other side. This proves the equivalence. Compute the finite graph list and apply Carter's algorithm: the required forbidden graph P_4 is explicitly present. □

An input family need not already be an antichain: redundant forbidden members can be removed by finite embedding tests, without changing the class. The theorem supplies a total decision procedure, conditional only on the published graph algorithm. We do not claim a new implementation or a new complexity bound for that algorithm.

Proposition 3.2 (Finite certificates for negative instances)**.** *If $\mathcal{C}(\mathcal{U})$ fails JEP, one can compute a pair F, G in the class and a finite exhaustive certificate that they have no common extension in the class.*

Proof. Any common host can be replaced by its induced suborder on the union of the two embedding images. Heredity preserves membership in the class, and the resulting host has at most $|F| + |G|$ vertices. For a fixed pair it therefore suffices to enumerate all forests of size at most that bound and test their membership and all candidate embeddings. These are finite calculations on finite order relations.

After a negative verdict from Theorem 3.1, enumerate permitted pairs and perform this finite test for each. A bad pair must eventually appear. The pair together with the exhaustive host test is a certificate verifiable without executing Carter's decision algorithm. □

The bound is a bound on the size of a host for a fixed pair. It is not a uniform bound on the size of the first bad pair as a function of the forbidden family. Without Carter's theorem the same search would semidecide only failure of JEP.

4 A sharp classification by width

The width of a finite poset is the largest size of an antichain. For a finite forest it is the number of maximal elements, which we call leaves. Indeed, each element of an antichain extends to a leaf, and incomparable elements have disjoint upper cones: a common upper bound would have incomparable predecessors.

Theorem 4.1. *For $m \geq 1$, the class of finite order forests of width at most m has JEP if and only if $m \leq 3$.*

Proof. First suppose $m \leq 3$. A forest of width at most m embeds as an induced suborder of a rooted binary tree with exactly m leaves. Add a root below all components; replace any multifurcation by binary branching through inserted vertices; and, if necessary, attach extra leaves. These operations preserve the old order and need at most m leaves. Suppressing

unary vertices now gives a full binary tree. For each of $m = 1, 2, 3$ there is exactly one unordered full binary tree with m leaves. A sufficiently long subdivision of each edge of this tree, together with a sufficiently long initial stem, therefore contains any prescribed pair of the original forests. Choose on each segment enough vertices for both individual embeddings; they need not have disjoint images. This common host still has width m.

Let $m \geq 4$. Compare a binary comb with m leaves, whose root split is $1 + (m-1)$, with a full binary tree whose root split is $2 + (m-2)$. For a vertex v let $L(v)$ be its set of descendant leaves. Distinct leaf clusters form a laminar family: two are nested or disjoint.

Suppose a full binary m-leaf tree embeds into a host H of width at most m. Its leaf images are incomparable; choose a host leaf above each one. There must be exactly m host leaves, and each source-leaf image has only its chosen host leaf above it. For a source vertex v, the host leaves above its image are precisely the matched leaves in $L(v)$. To check the nontrivial inclusion, any host leaf is above the image of a unique source leaf ℓ. If it is also above the image of v, foresthood makes those two images comparable. Order reflection then makes v and ℓ comparable, so $v \leq \ell$.

A full binary tree has $2m-1$ distinct nonempty clusters. This family is maximal laminar on the m leaves. Indeed, a set compatible with both children of every binary split must either be contained in one child or contain that entire child; descending recursively shows that it is already a cluster. The host's cluster family contains the source's and is laminar, so the two families coincide. Thus a common host for the two chosen sources would identify their cluster families under a leaf bijection. In the comb all nonsingleton proper clusters are nested. The other source has two disjoint nonsingleton root-child clusters. This is impossible. □

If A_k is the k-element poset antichain, then

$$\mathcal{C}(\{A_k\}) \text{ has JEP} \quad \Longleftrightarrow \quad k \leq 4 \qquad (k \geq 2).$$

For $m = 4$ the two negative witnesses each have seven vertices: the balanced binary tree and the four-leaf comb. Their embeddings were not assumed to preserve roots or meets. Cluster preservation was derived from the saturated width bound. If empty forests are allowed, the width-zero class consists only of the empty forest and has JEP too.

5 Countable realizations and scope

The age of a countable forest is the class of isomorphism types of its finite induced suborders. The following standard age construction explains one use of JEP without imposing amalgamation or homogeneity.

Proposition 5.1. *A nonempty hereditary class $\mathcal{K}$ of finite forests is the age of a finite or countable forest if and only if it has JEP. If it contains forests of unbounded size, the realizing forest is countably infinite.*

Proof. Any two finite suborders of one forest are contained in their finite union, proving necessity. Conversely enumerate the members of $\mathcal{K}$ up to isomorphism. Use JEP repeatedly to obtain a chain of members, with each enumerated member embedding into a later stage. Rename each successive host so that its chosen copy of the preceding stage is an actual induced suborder. The union is a forest: every finite violation of the order or predecessor-chain axioms would already occur at a finite stage. Every finite suborder of the union is contained in a stage, and hence belongs to $\mathcal{K}$ by heredity. Every member of $\mathcal{K}$ embeds into

some stage. The union therefore has exactly the required age. Unbounded size forces the union to be infinite. □

Together with Theorem 3.1, this decides which finite forbidden-family forest classes are ages. The empty class is treated separately: under the usual vacuous convention it has JEP, but is the age of no structure. Whether the empty forest is admitted as a structure is another convention that can be settled by a finite input check.

The source motivation in [2] concerns Scott analysis below the Vaught ordinal. The present paper answers its finite-forest Question 2 by an effective criterion, and gives a direct structural answer for antichain obstructions. It does not establish a model-counting theorem for arbitrary complete first-order theories. Nor does the age construction alone assign a Scott rank to its union. Such conclusions require the additional hypotheses of the relevant results in [2].

Preparation and status. This manuscript was assembled with AI assistance from a research programme directed by the repository owner. Carter's general algorithm is credited as a published input; no independent implementation or formal verification of it is claimed. The transfer and width proofs above are the arguments offered for review. No historical priority claim is made for those deductions. The repository contains the full manuscript and its build instructions; it requires no private project notes or conventions.

References

[1] D. Carter, *On the joint embedding property for cographs and trees*, arXiv:2409.06127v1 (2024), Corollary 1.5. `https://arxiv.org/abs/2409.06127v1`.

[2] D. Gonzalez, D. Rossegger and D. Turetsky, *Scott Analysis below the Vaught Ordinal*, arXiv:2606.15205v2 (2026), Question 2. `https://arxiv.org/abs/2606.15205v2`.